WORKSHEET

Chapter 12: Finding the gradient between two points on a line

The Gradient Formula: Gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

1 Find the gradient of the line between the following pairs of points.

a (1, 2) and (3, 6)

b (0,1) and (3, 4)

c (1, 0) and (2, 1)

d (2, 2) and (3, 6)

e (0, 0) and (2, 2)

f (0, 2) and (1, 0)

g (3, 5) and (4, 4)

h (1, 2) and (3, 3)

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i (0, 4) and (3, 6)

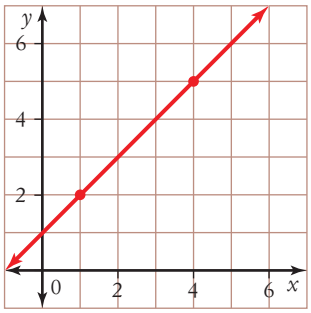
j (1, 1) and (5, 0)

k (3, 1) and (4, −2)

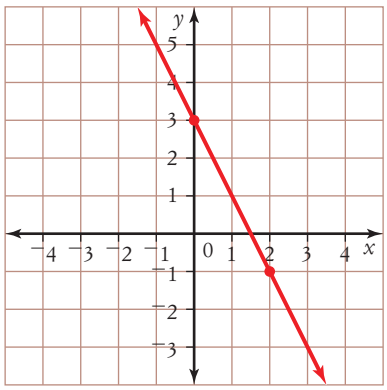
l (4, −2) and (−4, 2)

2 Write the coordinates of the two points marked on each line, then use them to calculate the gradient of the line.

a

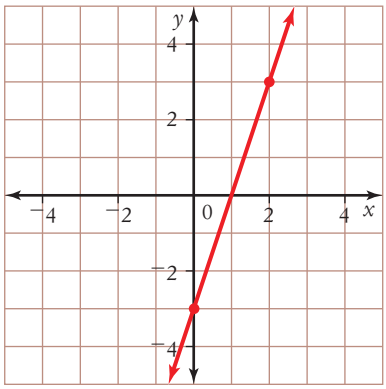


b

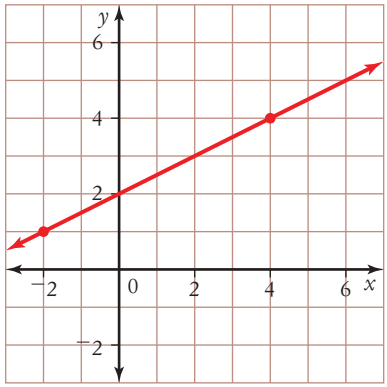


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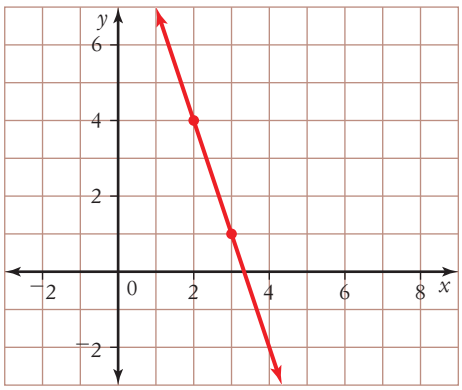
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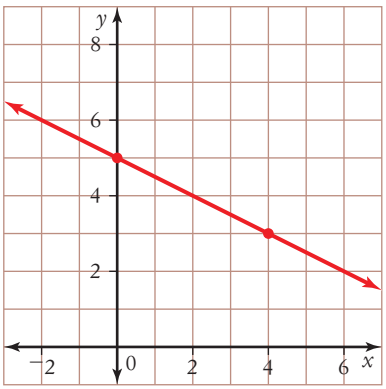
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e

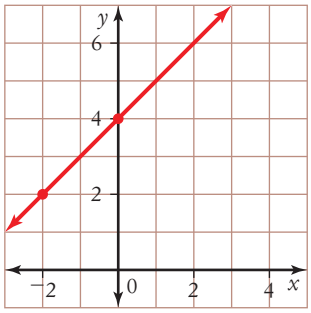


f

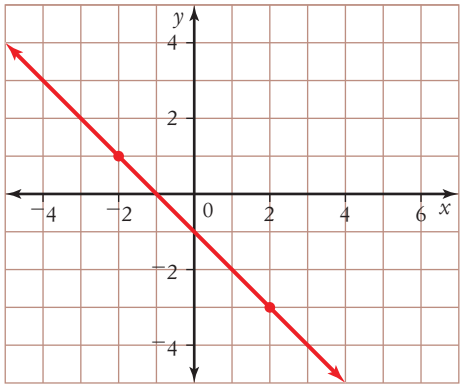


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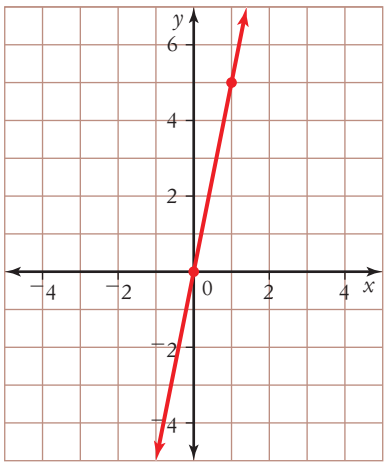
g



h



i



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3 Find the equation of each line, given the gradient and one point the line passes through.

a $b = 1, (2, 1)$

b $b = 2, (0, 1)$

c $b = -1, (2, 3)$

d $b = 3, (0, 2)$

e $b = 1, (-1, 0)$

f $b = -2, (3, 3)$

g $b = 4, (2, -1)$

h $b = 2, (-1, 1)$

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i $b = -1, (-3, 2)$

j $b = 0, (2, 4)$

k $b = \frac{1}{2}, (0, 0)$

l $b = -\frac{1}{2}, (2, 1)$

m $b = \frac{2}{3}, (6, 0)$

n $b = \frac{1}{2}, (4, 3)$

o $b = -\frac{1}{3}, (3, -1)$

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Answers

1 **a** 2
b 1
c 1
d 4
e 1
f -2
g -1
h $\frac{1}{2}$
i $\frac{2}{3}$
j $-\frac{1}{4}$
k -3
l $-\frac{1}{2}$

2 **a** $(1, 2), (4, 5), b = 1$
b $(0, 3), (2, -1), b = -2$
c $(0, -3), (2, 3), b = 3$
d $(-2, 1), (4, 4), b = \frac{1}{2}$
e $(2, 4), (3, 1), b = -3$
f $(0, 5), (4, 3), b = -\frac{1}{2}$
g $(-2, 2), (0, 4), b = 1$
h $(-2, 1), (2, -3), b = -1$
i $(0, 0), (1, 5), b = 5$

3 **a** $y = -1 + x$
b $y = 1 + 2x$
c $y = 5 - x$
d $y = 2 + 3x$
e $y = x + 1$
f $y = 9 - 2x$
g $y = -9 + 4x$
h $y = 3 + 2x$
i $y = -1 - x$
j $y = 4$
k $y = \frac{1}{2}x$
l $y = 2 - \frac{1}{2}x$
m $y = -4 + \frac{2}{3}x$
n $y = 1 + \frac{1}{2}x$
o $y = -\frac{1}{3}x$

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